

## Statistics for the Connected Components

1. Compute the total number of pixels in the connected component.
2. Compute the coordinates of the centroid pixel of the connected component computed as seen in figure 1.
3. Compute  $x_{\min}$ ,  $y_{\min}$ ,  $x_{\max}$  and  $y_{\max}$  for each connected component.  $x_{\min}$  and  $y_{\min}$  are the smallest x and y coordinates, respectively, of all of the pixels in the connected component.  $x_{\max}$  and  $y_{\max}$  are the largest x and y coordinates, respectively, of all of the pixels in the connected component. These four values form the coordinates of the smallest axis aligned rectangle that encompasses the connected component. Figure 2 illustrates the smallest axis aligned rectangle surrounding a connected component.

$$\left( \frac{\sum_{i=1}^j x_i}{j}, \frac{\sum_{i=1}^j y_i}{j} \right)$$

**Figure 1** - Coordinates of the centroid pixel of a component, where j is the number of pixels in the component.



**Figure 2** - The smallest axis aligned rectangle surrounding the character f.